

Coupling of plasmons to the spin-zero two-magnon continuum in antiferromagnets

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I. INTRODUCTION

Magnonics is an alternative approach to conventional computing, bypassing the Joule heating associated with electronic devices [1]. Within the field of magnonics, antiferromagnetic materials stand out from a technological perspective [2], due to their natural high frequencies in the Terahertz range. To further develop antiferromagnets as a materials platform for novel computing, it is of great importance to develop interoperable magnonics devices—to interface with the wide range of quasiparticles currently being investigated as possible information carriers in condensed matter, photonics and other fields.

One of the possible information carriers considered are plasmons, the quanta of charge density waves [3]. The emergence of two-dimensional materials that can support gapless plasmon modes, such as graphene [4–8], has opened up the possibility of magnon-plasmon coupling, and previous works have demonstrated this magnon-plasmon hybridization [9–12].

In this work, we show that plasmons can also couple to the spin-zero continuum formed by antiferromagnetic magnons, by coupling the electrically active plasmon through the electric polarization to the magnetic order [13]. We show that this could potentially realize a strong coupling between the plasmons and spin-zero two-magnon continuum, opening the way for coherent transfer of quantum information [14] between magnons and plasmons [15–17].

To demonstrate this coupling, we consider plasmons in a two-dimensional electron gas (2DEG), placed on top of an antiferromagnetic crystal, as shown in Fig. 1. The plasmons generate an electric field [18], which extends into the insulating antiferromagnetic crystal and couples to the magnetic excitations. This coupling has been previously considered in cuprates placed in a cavity [19] or driven by a laser [20]. However, magnonic energies in cuprates are much higher (in the 100 meV to 500 meV range) than magnonic energies in conventional antiferromagnets (in the 1 meV to 10 meV range), but cavities and lasers operating at these lower energy scales are much harder to realize. In this work, we therefore propose to use plasmons to bridge this gap.

In order for the magnetic excitations to couple to an electric field, the polarization operator in the antiferromagnet should be non-zero [13, 21]. In absence of spin-orbit coupling, it can be shown from symmetry considerations that this is only possible if the center of a bond connecting two

atoms is not a center of inversion. In this work we therefore choose an antiferromagnetic material where some (or all) bonds are not centers of inversion. We choose here two systems to illustrate the magnon-plasmon coupling, as shown in Fig. 1: (i) a Rutile-type antiferromagnet, where the anions naturally break the bond inversion symmetry, and (ii) a layered quasi two-dimensional antiferromagnet, where we break the bond inversion symmetry by placing non-magnetic ions in a checkerboard-like fashion.

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II. POLARIZATION

Microscopically, the resonant coupling between an electric field and the spin excitations in a magnetic material is described by the Hamiltonian [22]

$$\mathcal{H}_c = - \sum_i \hat{\mathbf{E}}_i \cdot \hat{\mathbf{P}}_i \quad (1)$$

where

$$\hat{\mathbf{P}}_i = \sum_j \left(\pi_{ij} \delta_{\alpha\beta} + \Gamma_{ij}^{(\alpha\beta)} + \mathbf{D}_{ij}^{[\alpha\beta]} \right) \hat{S}_i^\alpha \hat{S}_j^\beta \quad (2)$$

is the polarization operator expanded up to second order in spin operators. Here we have neglected the the polarization due to single spins, also known as the linear Stark effect, since the linear Stark effect vanishes if each lattice site is a centre of inversion, which we will assume to be the case here. We have decomposed the polarization tensor into the isotropic part π , the anisotropic traceless symmetric part Γ and the traceless antisymmetric part $\mathbf{D}$. The allowed components of this tensor are determined by lattice symmetries, and are modified in the presence of spin-orbit coupling. In this work we consider only the isotropic part π , which does not require spin-orbit coupling in antiferromagnets and is non-zero for any bond where its center is a center of inversion.

The direction of π is set by the crystal symmetries [13, 23], whereas its magnitude is dependent on the microscopic details. Generally, one can write $\pi_{ij} = g_{ij} a_{ij} (\mathbf{e}_{jk} - \mathbf{e}_{ki})$ [24], where g is a parameter describing the polarization amplitude, a_{ij} is the bond length and $\mathbf{e}_{ij}$ is the unit vector between atoms i and j . Within a Fermi-Hubbard model at half filling it is possible to derive the polarization amplitude as $g_{ij} = 8et_{ij}t_{jk}t_{ki}/U^3$, where e is the electron charge, t_{ij} is the hopping amplitude between i and j , and U is the Coulomb repulsion [24]. Note that this expression was derived in the Mott insulating limit, and therefore $t/U \ll 1$. Not all antiferromagnets are accurately described by this model, and other processes can also

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contribute to the polarization amplitude. In this work, we will therefore consider the polarization amplitude g_{ij} to be a phenomenological parameter, proportional to the bond length.

We continue by considering the electric field operator $\hat{E}_i$ of a plasmon in the 2DEG, which at position $\mathbf{r}_i$ in the antiferromagnetic film is given by

$$\hat{E}(\mathbf{r}_i) = \frac{1}{\sqrt{N_p}} \sum_q i\mathbf{E}_q e^{iq \cdot \mathbf{r}_i - |z_i|/\lambda_q} \hat{\phi}_q + \text{h.c.} \quad (3)$$

with $\hat{\phi}_q$ the plasmon annihilation operator and $\mathbf{E}_q(z)$ a mode function, which captures the details of the plasmonic dispersion. We give $\mathbf{E}_q(z)$ and further details of the plasmons in Appendix D. Here N_p is the number of atomic sites in the 2DEG and the plasmon electric field decays exponentially along the z -axis, with a decay length $\lambda^{-1} \approx q$.

Continuing with only the isotropic part π of the polarization tensor, the coupling Hamiltonian can then be written as

$$\mathcal{H}_c = - \sum_{ij} (\hat{E}(\mathbf{r}_i) \cdot \pi_{ij}) \hat{S}_i \cdot \hat{S}_j, \quad (4)$$

where we have assumed that the plasmon electric field does not modulate over atomic distances, such that $\langle \mathbf{E}(\mathbf{r}_i) \rangle \approx \langle \mathbf{E}(\mathbf{r}_j) \rangle$ for neighboring pairs $\langle ij \rangle$.

III. METHOD

We consider here an antiferromagnet described by the Heisenberg Hamiltonian,

$$\mathcal{H}_{\text{AFM}} = \sum_{ij} J_{ij} \hat{S}_i \cdot \hat{S}_j - \frac{K}{2} \sum_i (\hat{S}_i^z)^2, \quad (5)$$

where J_{ij} is the Heisenberg exchange strength between spins at site i, j and $K > 0$ parameterizes an easy-axis anisotropy. We assume that the ground state of the system is antiferromagnet, such that its elementary magnetic excitations are magnons. To describe the magnonic excitations, we employ a Holstein-Primakoff transformation, apply a Fourier transformation and a Bogoliubov transformation [see Appendix A for the details], such that the Hamiltonian Eq. (5) becomes

$$H = \sum_k \epsilon_k^+ \hat{\alpha}_k^\dagger \hat{\alpha}_k + \epsilon_k^- \hat{\beta}_k^\dagger \hat{\beta}_k, \quad (6)$$

where $\epsilon_k^\pm$ are the magnon eigenenergies.

We will consider in this work two examples of antiferromagnet with non-zero electric-dipole coupling: (i) a three-dimensional antiferromagnet and (ii) a quasi-two dimensional layered antiferromagnet, as shown in Fig. 1. As a model for the 3D AFM we take an easy-axis Rutile antiferromagnet, such as MnF_2 or FeF_2 , where the placement of the anions removes the center of inversion located between neighboring ions. For the quasi-2D AFM we take inspiration from the cuprates and consider layers of a square lattice AFM, with an additional non-magnetic ion placed in a checkboard fashion—which removes the center of inversion located between neighboring magnetic ions [20]. Both compounds retain a global

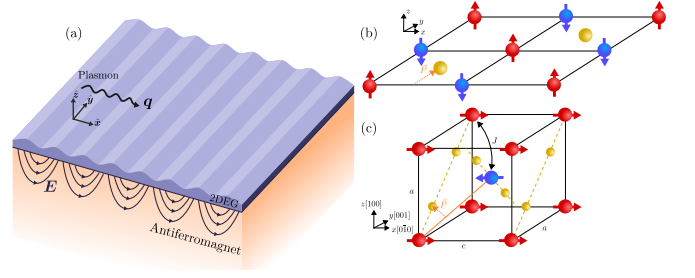


FIG. 1. (a) The setup considered. (b) A quasi-two dimensional antiferromagnet. The magnetic ions (red and blue) are antiferromagnetically ordered and form a square lattice, and the non-magnetic ions (gold) break the center of inversion symmetry located between neighboring magnetic ions. This allows for a finite polarization vector $\vec{P}$, as indicated for one bond. (c) A three-dimensional antiferromagnet inspired by the Rutile-type antiferromagnets, such as MnF_2 , grown along [100]. The magnetic ions are antiferromagnetically ordered, and the anions break the inversion symmetry along the nearest-neighbor bonds. For one such bond the direction of the polarization vector is indicated. **PG: double check the orientation of the Rutile**

inversion symmetry. We consider a semi-infinite bulk crystal, which for the Rutile-based AFM can be treated in the three-dimensional limit, whereas the cuprate-inspired quasi-2D AFM can be treated as decoupled 2D layers.

A. Magnon-plasmon coupling

We now assume that some (or all) bonds are not centers of inversion, such that the isotropic part of the polarization operator, π_{ij} , is non-zero. We introduce the Holstein-Primakoff operators in the coupling Hamiltonian Eq. (4), to find, up to second order in Bogoliubov-transformed Holstein-Primakoff operators, the coupling Hamiltonian

$$H_c^{(2)} = -iS \frac{1}{\sqrt{N_x N_y}} \sum_n e^{-|z_n|/\lambda_q} \sum_{\mathbf{k}q} (\hat{\phi}_q \mathbf{E}_q - \hat{\phi}_{-q}^* \mathbf{E}_{-q}^*) \cdot \hat{\mathbf{V}}_{\mathbf{k},q} \quad (7)$$

where $\hat{\mathbf{V}}_{\mathbf{k},q} \equiv \hat{\mathbf{V}}_{\mathbf{k},q}^{\text{pair}} + \hat{\mathbf{V}}_{\mathbf{k},q}^{\text{scatt}}$, with

$$\begin{aligned} \hat{\mathbf{V}}_{\mathbf{k},q}^{\text{pair}} \equiv & u_{\mathbf{k}} u_{\mathbf{k}-q} \pi_{-q-k}^* \hat{\alpha}_{\mathbf{k}} \hat{\beta}_{\mathbf{k}-q} \\ & + u_{\mathbf{k}} u_{\mathbf{k}+q} \pi_{q-k} \hat{\alpha}_{\mathbf{k}}^\dagger \hat{\beta}_{\mathbf{k}+q}^\dagger \\ & + v_{\mathbf{k}} v_{\mathbf{k}-q} \pi_{q+k} \hat{\alpha}_{-k-q} \hat{\beta}_{\mathbf{k}} \\ & + v_{\mathbf{k}} v_{\mathbf{k}+q} \pi_{k-q}^* \hat{\beta}_{\mathbf{k}}^\dagger \hat{\alpha}_{q-k}^\dagger \end{aligned} \quad (8)$$

containing the pair creation and $\hat{\mathbf{V}}_{\mathbf{k},q}^{\text{scatt}}$ describes the scattering processes, containing terms $\propto \hat{\alpha}_{\mathbf{k}} \hat{\alpha}_{\mathbf{k}+q} + \text{h.c.}$ and $\propto \hat{\beta}_{\mathbf{k}} \hat{\beta}_{\mathbf{k}+q} + \text{h.c.}$. Furthermore, $u_{\mathbf{k}}$ and $v_{\mathbf{k}}$ are the Bogoliubov factors defined in Appendix A. Here we have assumed that for every in-plane magnetic sublattice we have one atomic site in the 2DEG, and that the plasmon electric field, $\mathbf{E}_q(z_n)$ is slowly varying along the z axis, such that the z -dependency can be

considered as a slowly varying potential. Finally,

$$\pi_{\mathbf{k}} \equiv \sum_{ij} e^{-i\mathbf{r}_{ij} \cdot \mathbf{k}} \pi_{ij} \quad (9)$$

is the Fourier transform of the isotropic part of the polarization tensor.

For a Rutile AFM, the anions breaks the inversion symmetry along specific bonds, thus allowing for a finite polarization, as indicated for a single bond in Fig. 1. This results in the polarization

$$\pi_{\mathbf{k}}^R = ig_R \begin{pmatrix} 1.63a \cos(\frac{1}{2}k_x a) \cos(\frac{1}{2}k_y c) \cos(\frac{1}{2}k_z a) \\ 7.63c \sin(\frac{1}{2}k_x a) \sin(\frac{1}{2}k_y c) \sin(\frac{1}{2}k_z a) \\ 1.63a \sin(\frac{1}{2}k_x a) \cos(\frac{1}{2}k_y c) \cos(\frac{1}{2}k_z a) \end{pmatrix}. \quad (10)$$

For the two-dimensional AFM we have that the bond inversion symmetry is broken by the non-magnetic ions placed in a checkerboard, which gives

$$\pi_{\mathbf{k}}^{2D} = ig_{2D} \begin{pmatrix} 2a \sin(k_y a) \\ 2a \sin(k_x a) \\ 0 \end{pmatrix}. \quad (11)$$

Here g_R, g_{2D} are a phenomenological parameter with units charge.

This concludes the coupling of the plasmons and the magnons. We can directly write down the resulting plasmon self-energy [25] up to second order in the electric field, of which there are three kinds, depicted in Fig. 3: (a) forward, (b) backward and (c) circle diagrams. Since magnons are non-conserved bosonic particles, at zero temperature no magnons are present and thus the circle diagram does not contribute. Since the scattering processes only enter in the circle diagram, we can neglect them from here on out.

The main contribution will come from the forward diagram, which which we can directly write the plasmon self-energy at zero temperature as

$$\Sigma_q^f(\epsilon) = \frac{1}{N_x N_y} \sum_n e^{-|z_n|/\lambda_q} \sum_{\mathbf{k}} \sum_{\sigma, \sigma'=\pm} \frac{1}{2} \frac{|\mathbf{E}_q \cdot \mathbf{V}_{q, \mathbf{k}}^{\sigma\sigma'}|^2}{\epsilon + i0^+ - \epsilon_{\mathbf{k}}^{\sigma} - \epsilon_{\mathbf{q}-\mathbf{k}}^{\sigma'}}. \quad (12)$$

Here $\mathbf{V}_{q, \mathbf{k}}^{\sigma\sigma'}$ is the interaction vertex in the Boguliobov basis, given by

$$\mathbf{V}_{q, \mathbf{k}}^{\sigma\sigma'} \equiv S \begin{pmatrix} 0 & u_{\mathbf{k}} u_{\mathbf{k}+\mathbf{q}} \pi_{-\mathbf{q}+\mathbf{k}} \\ v_{\mathbf{k}} v_{\mathbf{k}+\mathbf{q}} \pi_{-\mathbf{q}+\mathbf{k}} & 0 \end{pmatrix}. \quad (13)$$

Here we note that $u_{\mathbf{k}} u_{\mathbf{k}+\mathbf{q}}$ and $v_{\mathbf{k}} v_{\mathbf{k}+\mathbf{q}}$ reduce to the usual exchange enhancement factors for $q \ll k$, which is typically the case given the mismatch in wavelength between magnons and plasmons. The backward diagram gives

$$\Sigma_q^b(\epsilon) = \frac{-1}{N_x N_y} \sum_n e^{-|z_n|/\lambda_q} \sum_{\mathbf{k}} \sum_{\sigma, \sigma'=\pm} \frac{1}{2} \frac{|\mathbf{E}_q \cdot \mathbf{V}_{-q, \mathbf{k}}^{\sigma\sigma'}|^2}{\epsilon + i0^+ + \epsilon_{\mathbf{k}}^{\sigma} + \epsilon_{\mathbf{q}-\mathbf{k}}^{\sigma'}}, \quad (14)$$

and is purely real, and is thus less significant for the plasmon self-energy, which is defined as

$$\Sigma_q(\epsilon) \equiv \Sigma_q^f(\epsilon) + \Sigma_q^b(\epsilon). \quad (15)$$

In addition, we find numerically that typically $\Sigma_q^b(\epsilon) \ll \text{Re}[\Sigma_q^f(\epsilon)]$, and we thus discuss only the forward diagram here. In the numerical result that follow, we do however include the backward diagram for completeness.

The self energy in Eq. (12) is valid in two and three dimensions, and takes into account the decay of the plasmon electric field. We now assume that the antiferromagnet is much thicker than the plasmon decay length, such that we can take $N_z \rightarrow \infty$ and explicitly perform the summation over n to find $\sum_n e^{-|z_n|/\lambda_q} = (e^{2a_z/\lambda_q} - 1)^{-1}$, where a_z is the lattice constant along the z axis.¹ This scaling corresponds to the fact that a large number of magnon modes are available for a plasmon mode to decay into, which is only limited by the decay length of the plasmon electric field. This is similar to how magnon-photon coupling in a cavity is proportional to square root of the number of exchange-coupled spins [26].

We observe that the spin-zero two-magnon density of states (DOS) defined as

$$D_q(\epsilon) = \frac{1}{N_m} \sum_{\substack{\sigma, \sigma'=\pm \\ \sigma \neq \sigma'}} \sum_{\mathbf{k}} \delta(\epsilon - \epsilon_{\mathbf{k}}^{\sigma} - \epsilon_{\mathbf{q}-\mathbf{k}}^{\sigma'}) \quad (16)$$

is directly related to the imaginary part of the plasmon self-energy,

$$\text{Im}[\Sigma_q(\epsilon)] \propto \frac{1}{N_x N_y} \sum_{\mathbf{k}} \sum_{\sigma, \sigma'=\pm} \frac{1}{2} |\mathbf{E}_q \cdot \mathbf{V}_{q, \mathbf{k}}^{\sigma\sigma'}|^2 \delta(\epsilon - \epsilon_{\mathbf{k}}^{\sigma} - \epsilon_{\mathbf{q}-\mathbf{k}}^{\sigma'}), \quad (17)$$

weighed by the interaction vertex. The spin-zero two-magnon DOS therefore plays an important role in determining the plasmon self-energy.

We furthermore note that the higher-order magnon-magnon interactions will renormalize the plasmon self-energy, since the excited magnon pair are close to each other in real space and will thus interact through magnon-magnon interactions. The main effect of this interaction is to cut the Van Hove singularity at the Brillouin zone edge. We take this broadening into account by introducing a finite magnon lifetime, $\epsilon_{\mathbf{k}}^{\sigma} \rightarrow \epsilon_{\mathbf{k}}^{\sigma}(1 - i\alpha)$, where $\alpha = 10^{-3}$ is chosen to reproduce the broadening calculated through many-body theory in the context of Raman scattering [27]. Note that the magnon-magnon interaction will also induce a small energy shift, which we neglect here.

The plasmon self-energy is derived perturbatively, and thus assumes that the magnon-plasmon coupling is small. This is however not a straightforward requirement, since the plasmon self-energy is strongly enhanced due to the Van Hove singularity at the zone boundary. We therefore additionally require that the plasmon self-energy is smaller than the plasmon energy, i.e., $|\Sigma_q| < \epsilon$. We enforce this requirement by choosing a suitably small coupling constant g . Beyond the perturbative limit, i.e., if $|\Sigma_q| > \epsilon$, this system should instead be treated in the strong-coupling regime [28].

¹ For a thin film thinner than the decay length, one instead obtains $(1 - e^{-2a_z N_z/\lambda_q})/(e^{2a_z/\lambda_q} - 1)$.

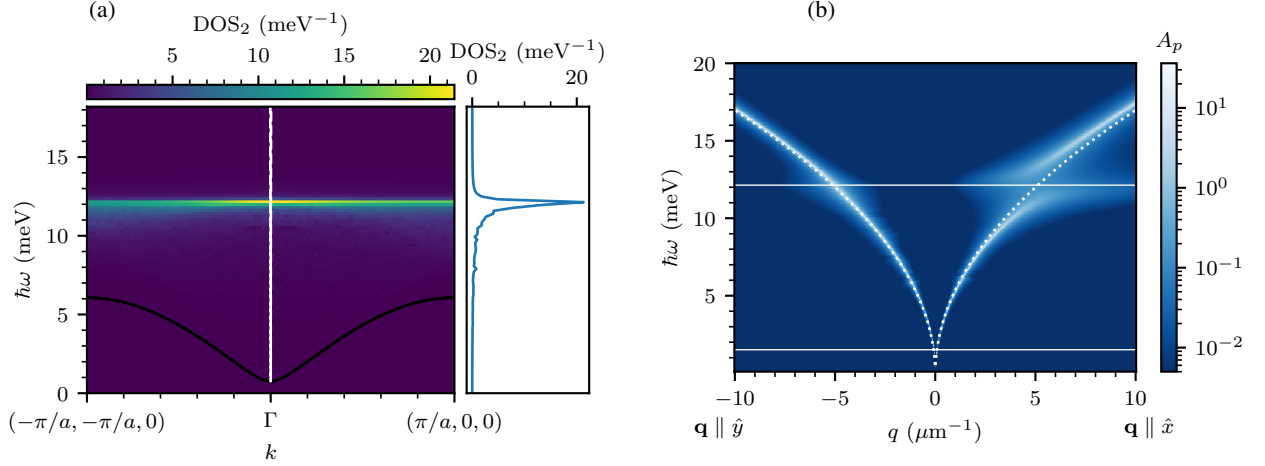


FIG. 2. For a Rutile-type antiferromagnet: (a) the magnon dispersion in the $kx - k_y$ plane. The dashed white line indicate the plasmon dispersion and the colorscale is the spin-zero two-magnon density of states. Shown on the right is a cut of the spin-zero two-magnon density of states along the plasmon energy. Horizontal dashed lines indicate the upper and lower edges of the spin-zero two-magnon continuum. (b) The plasmon spectral function $A_q(\epsilon)$ along two propagation directions of the plasmon. The lower and upper edges of the spin-zero two-magnon continuum are indicated by the horizontal white lines. The bare plasmon dispersion is indicated by the dashed white line.

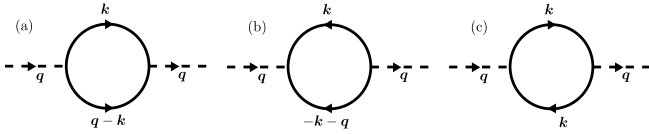


FIG. 3. Feynmann diagrams contributing to the plasmon lifetime up to second order. (a) Forward, (b) backward and (c) circle diagram. At zero temperature, only the forward and backward diagrams contribute.

IV. RESULTS

We now analyze the renormalized plasmon spectrum, by calculating the spectral function [25]

$$A_q(\epsilon) = -\frac{1}{\pi} \text{Im}[\mathcal{G}_q(\epsilon)], \quad (18)$$

where

$$\mathcal{G}_q(\epsilon) = [\epsilon + i0^+ - \mathcal{E}_q(1 - i\gamma) - \Sigma_q(\epsilon)]^{-1} \quad (19)$$

is the plasmon retarded Greens function. Here $\mathcal{E}_q \propto \sqrt{q}$ is the plasmon dispersion, which we give in Appendix D. We have included a finite plasmon lifetime through the inverse plasmon quality factor, which we set to $\gamma = 10^{-2}$ [29]. Furthermore, we set $g_R = 10^{-3} \times e$ and $g_c = 5 \times 10^{-3} \times e$, which is comparable in magnitude to the polarization amplitude calculated for Sr_2IrO_4 [23] and Cr_2O_3 [30] and for a Hubbard model with $t/U = 0.1$ [24]. These polarization amplitudes were chosen to remain in the perturbative regime, i.e., such that we have $|\Sigma_q| < \epsilon$. For simplicity, we also consider MnF_2 to be cubic, i.e., $a = c$.

A. Rutile-based antiferromagnet

We show the resulting plasmon spectral function for a Rutile-based AFM in Fig. 2(a), as a function of wave vector q and energy ϵ , for a plasmon propagating along x and y . We have chosen the wave vector range such that the plasmon energies are comparable to the magnon energies. The white dashed line indicates the bare plasmon dispersion. The lower and upper edges of the spin-zero two-magnon continuum are indicated by the horizontal white lines. For the range of wave vectors considered here, the decay length λ is comparable to the wave length, i.e., of the order of several micrometers, which will significantly enhance the coupling.

We first focus on a plasmon propagating along x , where we observe that below the spin-zero two-magnon continuum the plasmon dispersion is unaffected. Upon entering the spin-zero two-magnon continuum, the plasmon decays into the continuum [28], in particular at the singularity at the upper edge of the continuum. After exiting the spin-zero two-magnon continuum, the plasmon becomes visible again, but its dispersion is normalized—as can be seen by comparing the plasmon spectral function above the spin-zero two-magnon continuum with the base dispersion (white dashed line).

Here we stress that we remain in the weak-coupling regime, such that $|\Sigma_q(\epsilon)| < \epsilon$ for all ϵ . Since the polarization strength g enters quadratically, it is however relatively easy to reach the strong-coupling regime, where $|\Sigma_q(\epsilon)| > \epsilon$, for slightly stronger polarization strengths. As was also explained before, here our perturbative treatment breaks down and we are unable to reliably calculate the full plasmon spectral function. We have thus restricted ourselves here to the weak-coupling regime, but it seems likely that the strong-coupling regime can also be reached. In the strong-coupling regime, it might also be possible to observe the expulsion of the plasmon from the

Van Hove singularity [28].

We next turn to a plasmon along y , where the coupling to the continuum is much weaker, and there is no noticeable decay. This can be explained by the electric field of the plasmon, which is parallel to its propagation direction, in addition to an out-of-plane electric field, i.e., $\mathbf{E} = E_{\parallel}\hat{q} + E_z\hat{z}$ [Appendix D]. Thus, a plasmon along y only couples to the y and z component of the polarization vector, π_k [Eq. (10)]. Because of the Van Hove singularity at the Brillouin zone boundaries, we furthermore have that mainly magnons at the Brillouin zone boundaries contribute to the plasmon self-energy. However, the x -component of the polarization vanishes at the zone boundaries, whereas the y -component of the polarization has a maximum there, and the z -component of the polarization has a maximum only along the $k_x = \pm\pi/a$ edges of the Brillouin zone. A plasmon propagating along y therefore couples much stronger to the spin-zero two-magnon continuum. Here we note that the symmetries of π_k are set by the crystal symmetry, and could thus change in different compounds—offering a potential way to change the strength of the magnon-plasmon coupling. In particular, for the Rutiles grown along the [001]-axis, π_k inherits the $PG: ???$ symmetry of the lattice and thus the self energy is equivalent along x and y .

The imaginary part of the plasmon self-energy is only non-zero if the spin-zero two-magnon density of states, Eq. (16), is finite. To explore this in more detail, we show the spin-zero two-magnon continuum in Fig. 2(b). We also show the spin-zero two-magnon DOS at the at $k = \Gamma$ in the right panel. We first note that due to the wavelength mismatch between magnons and plasmons in a Rutile AFM, the plasmons have effectively zero wave length compared to the magnons. The two-magnon DOS contains two analytic discontinuities: upon entering the continuum, at $\epsilon = 2K$, and upon exiting the continuum at $\epsilon = 2K + 2zJ$. In three dimensions these discontinuities take the form of a square root. Secondly, the spin-zero two-magnon DOS peaks strongly at the upper edge of the spin-zero two-magnon continuum, where a Von Hove singularity is located, corresponding to the excitation of two zone edge boundaries, where the magnon dispersion is flat. These two features of the spin-zero two-magnon DOS explain why the spin-zero two-magnon continuum repels the plasmon, and why this repelling originates from the upper edge of the spin-zero two-magnon continuum. Note here that the the Van Hove singularity is not a true singularity, but is cut by the magnon lifetimes [31] and higher order magnon interactions [27], which we have modeled here by introducing a finite magnon lifetime. We also stress that at the Brillouin zone edge, the the exchange enhancement factors tend to unity, $u_k^2 \approx v_k^2 \approx 1$. In the plasmon self-energy we thus do not see a significant exchange enhancement.

B. Quasi-two dimensional antiferromagnet

We now consider the quasi-2D AFM, and show the resulting plasmon spectral function in Fig. 4, considering the propagation of plasmons along $\hat{x}$ and $\hat{y}$. As before, the white dashed line indicates the bare plasmon dispersion and the lower and

upper edges of the spin-zero two-magnon continuum are indicated by the horizontal white lines.

From the plasmon spectral function in Fig. 4(b) we observe that the plasmon again strongly decays as it enters the Van Hove singularity in the spin-zero two-magnon DOS, similar to the results obtained for the Rutile AFM. Above the continuum, the plasmon is again renormalized. This therefore qualitatively reproduces the results for the three-dimensional Rutile-based AFM. Note again that we have restricted ourselves to the weak-coupling regime.

Note however that the polarization is invariant under rotation in the plane, and thus the plasmon self-energy is identical for all propagation directions—as can also be seen by comparing the propagation along x and y in Fig. 4(b).

Finally, we note that in two dimensions the discontinuities in the imaginary part of the plasmon self-energy upon entering and exiting the spin-zero two-magnon continuum is a step function, which implies a logarithmic divergence in the real part [32]. This could potentially enhance the magnon-plasmon interaction compared to the three-dimensional Rutile AFM, but this divergence is smeared out by the magnon-magnon interactions [27]. Furthermore, at the lower edge of the spin-zero two-magnon continuum, the continuum is formed of two magnons with (almost) zero momentum, for which the polarization, Eq. (11), vanishes, and thus the plasmon self-energy is zero—which we confirmed numerically (not shown here).

V. CONCLUSION AND DISCUSSION

We have shown in this work that plasmons couple to the spin-zero two-magnon continuum in antiferromagnetic thin films with a finite bond polarization. This allows a plasmon to excite and annihilate magnon pairs and thus couple to the spin-zero two-magnon continuum. For weak coupling, the plasmon mode decays into the spin-zero two-magnon continuum, whereas for sufficiently strong coupling, the plasmon is expelled from the spin-zero two-magnon continuum. We predict that the strong coupling regime could potentially be achieved in Rutile and quasi-two-dimensional antiferromagnets.

Here one could reasonably ask the question if the strong coupling is not due to a failure of the perturbation theory used to obtain the self-energy expressions. Firstly, we note that we have neglected here the magnon self-energies which also arise from the same coupling Hamiltonian Eq. (7). We show however in Appendix C that the decay of a magnon into a plasmon and a magnon is kinematically not allowed, due to the strong mismatch in wavelength between the magnons and plasmons—and thus the magnon self-energy vanishes. Up to second order in the plasmon electric field $\mathbf{E}_q$ our theory is therefore complete. This leaves diagrams which are of higher order in $\mathbf{E}_q$ to potentially renormalize the coupling of the plasmon to the spin-zero two-magnon continuum. However, since $\mathbf{E}_q$ is typically small, we do not expect this effect to be significant. This leaves the quadratic magneto-optical couplings which are typically considered in two-magnon Ra-

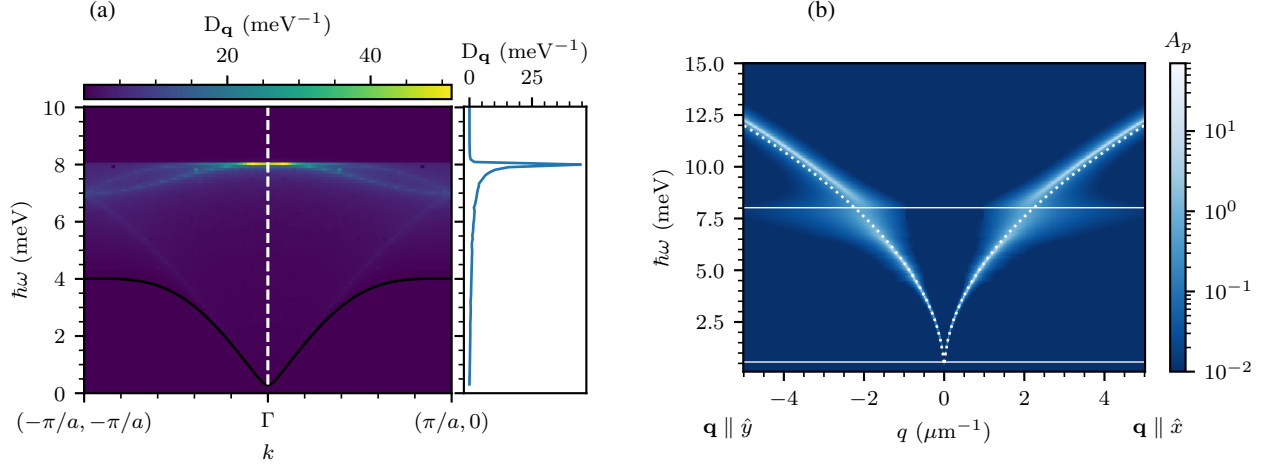


FIG. 4. For a quasi two-dimensional antiferromagnet: (a) the magnon dispersion. The dashed white line indicate the plasmon dispersion and the colorscale is the spin-zero two-magnon density of states. Shown on the right is a cut of the spin-zero two-magnon density of states along the plasmon energy. Horizontal dashed lines indicate the upper and lower edges of the spin-zero two-magnon continuum. (b) The plasmon spectral function $A_q(\epsilon)$ along two propagation directions of the plasmon. The lower and upper edges of the spin-zero two-magnon continuum are indicated by the horizontal white lines. The bare plasmon dispersion is indicated by the dashed white line.

man scattering, which are quadratic in plasmon operators. We show in Appendix C that these self energies have a reduced phase space available and we thus expect this self energy to be significantly weaker.

We have considered here an antiferromagnetic film thick enough to be treated in the three-dimensional limit. We note here that it is possible to treat the antiferromagnetic film as three dimensional if the magnons participating in the plasmon self energy have a wavelength shorter than the film thickness. Since mainly Brillouin zone edge magnons participate in the self energy due to the Van Hove singularity located there, we expect that this condition is satisfied even for films down to tens of layers.

Within the AFMs models considered in this work, the polarization strength, g , is a critical parameter to strongly couple the plasmon to the spin-zero two-magnon continuum. We note here that several antiferromagnetic materials exist, such as multi-orbital magnetic Mott insulators [13], Sr_2IrO_4 [23] and Cr_2O_3 [30]. Additionally, cuprates are a good candidate [19, 20], and are typically well-described by the Fermi-Hubbard model, which allows one to calculate the polarization [24]. However, we note here that the large magnon bandwidth (in the order of 100s of meV) might prohibit the coupling of plasmons to the spin-zero two-magnon continuum, since typical plasmons energies do not extend to such large values. Alternatively, it could be possible to engineer higher-energy plasmons, such as by making use of interband plasmons in moiré graphene-metal heterostructures [33].

Another exciting future direction is the coupling of plasmons to multiferroics, where the magnetoelectric effects can be large [34, 35]. In a multiferroic the magnons hybridize with the electric polarization, and thus the low-lying modes are electromagnons [36–38]. The finite polarization of the electromagnons could potentially enhance their coupling to

the plasmons.

Appendix A: Antiferromagnetic magnon Hamiltonian

We introduce the Holstein-Primakoff operators

$$\hat{S}_{A;i}^+ \approx \sqrt{2S} \hat{a}_i; \quad \hat{S}_{A;i}^z = S - \hat{a}_i^\dagger \hat{a}_i; \quad (\text{A1})$$

$$\hat{S}_{B;i}^+ \approx \sqrt{2S} \hat{b}_i^\dagger; \quad \hat{S}_{B;i}^z = -S + \hat{b}_i^\dagger \hat{b}_i \quad (\text{A2})$$

in the Heisenberg Hamiltonian Eq. (5) and apply the Fourier transformation, $\hat{a}_i = \frac{1}{\sqrt{N_m}} \sum_{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}_i} \hat{a}_{\mathbf{k}}$, where $N_m = N_x \times N_y \times N_z$ is the number of spins in each sublattice, to find the Hamiltonian for $\eta \in \{\text{Rutile}, 2D\}$,

$$\begin{aligned} \hat{H}_{\text{AFM};\eta}^{(2)} = \sum_{\mathbf{k}} & (zJ + K + J'_{\eta;\mathbf{k}}) \hat{a}_{\mathbf{k}}^\dagger \hat{a}_{\mathbf{k}} + (zJ + K + J'_{\eta;\mathbf{k}}) \hat{b}_{\mathbf{k}}^\dagger \hat{b}_{\mathbf{k}} \\ & + J_{\eta;\mathbf{k}} (\hat{a}_{\mathbf{k}} \hat{b}_{-\mathbf{k}} + \hat{a}_{\mathbf{k}}^\dagger \hat{b}_{-\mathbf{k}}^\dagger) \end{aligned} \quad (\text{A3})$$

with

$$J_{\text{Rutile};\mathbf{k}} = zJ \cos(\frac{1}{2}k_x a) \cos(\frac{1}{2}k_y a) \cos(\frac{1}{2}k_z a) \quad (\text{A4})$$

$$J_{2D;\mathbf{k}} = \frac{1}{2} zJ [\cos(k_x a) + \cos(k_y a)] \quad (\text{A5})$$

and

$$J'_{\text{Rutile};\mathbf{k}} = 0 \quad (\text{A6})$$

$$J'_{2D;\mathbf{k}} = zJ' \cos(k_x a) \cos(k_y a) \quad (\text{A7})$$

This Hamiltonian can be diagonalized through the Bogoliubov transformation,

$$\begin{aligned} \hat{a}_{\mathbf{k}} &= u_{\mathbf{k}} \hat{\alpha}_{\eta;\mathbf{k}} - v_{\eta;\mathbf{k}} \hat{\beta}_{-\mathbf{k}}^\dagger \\ \hat{b}_{-\mathbf{k}}^\dagger &= -v_{\eta;\mathbf{k}} \hat{\alpha}_{\mathbf{k}} + u_{\eta;\mathbf{k}} \hat{\beta}_{-\mathbf{k}}^\dagger \end{aligned} \quad (\text{A8})$$

to find

$$\hat{H}_\eta = \sum_{\mathbf{k}} \epsilon_{\eta;\mathbf{k}} (\hat{\alpha}_{\mathbf{k}}^\dagger \hat{\alpha}_{\mathbf{k}} + \hat{\beta}_{\mathbf{k}}^\dagger \hat{\beta}_{\mathbf{k}}) \quad (\text{A9})$$

with

$$\epsilon_{\mathbf{k}} = \sqrt{(zJ_\eta + K + J'_{\eta;\mathbf{k}})^2 - J_{\eta;\mathbf{k}}^2} \quad (\text{A10})$$

as the antiferromagnetic dispersion relation and

$$u_{\eta;\mathbf{k}} = \cosh \varphi_{\eta;\mathbf{k}}; \quad v_{\eta;\mathbf{k}} = \sinh \varphi_{\eta;\mathbf{k}} \quad (\text{A11})$$

the Bogoliubov factors, with **PG: check!**

$$\tanh(2\varphi_{\eta;\mathbf{k}}) \equiv \frac{-J_{\eta;\mathbf{k}}}{zJ_\eta + K + J'_{\eta;\mathbf{k}}}. \quad (\text{A12})$$

Throughout this work, we will set $S = 1$. For the Rutile AFM, we choose the parameters for MnF_2 , $J = 0.75$ meV, $K = 0.05$ meV, [39] and for the quasi-2D AFM, we take $J = 1.0$ meV, $K = 0.01$ meV. Note that in the Rutile-AFM, the anions could introduce a non-zero second-nearest neighbor coupling, which leads to an alternating splitting of the band structure, but this splitting is negligible for MnF_2 and most other insulating cuprates [40].

Appendix B: Magnon-plasmon coupling

We insert the Holstein-Primakoff operators in the coupling Hamiltonian Eq. (4) to find

$$H_c^{(2)} = -S \sum_{ij} (\hat{\mathbf{E}}(\mathbf{r}_i) \cdot \boldsymbol{\pi}_{ij}) (\hat{a}_i \hat{b}_j + \hat{a}_i^\dagger \hat{b}_j^\dagger), \quad (\text{B1})$$

where we have discarded constant energy shifts and $\hat{a}_i^\dagger \hat{a}_i + \hat{b}_i^\dagger \hat{b}_i$, since $\boldsymbol{\pi}_{ii} = 0$. We assume that for every in-plane magnetic sublattice we have one atomic site in the 2DEG, such that $N_p = N_x \times N_y$. We Fourier transform the Holstein-Primakoff operators to find

$$H_c^{(2)} = -iS \sqrt{\frac{N_z}{N_m}} \sum_{\mathbf{k}q} (\hat{\phi}_q \mathbf{E}_q - \hat{\phi}_{-q}^* \mathbf{E}_{-q}^*) \cdot \left[\boldsymbol{\pi}_{-q-k}^* \hat{a}_k \hat{b}_{-q-k} + \boldsymbol{\pi}_{q-k} \hat{a}_k^\dagger \hat{b}_{q-k}^\dagger \right] \quad (\text{B2})$$

where

$$\boldsymbol{\pi}_{\mathbf{k}} \equiv g \sum_{ij} e^{-i\mathbf{r}_{ij} \cdot \mathbf{k}} \boldsymbol{\pi}_{ij} \quad (\text{B3})$$

is the Fourier transform of the isotropic part of the polarization tensor, given in Eqs. (10) and (11) for the Rutile and quasi-2D AFM. Here we have written $1/\sqrt{N_p} = 1/\sqrt{N_x \times N_y} = \sqrt{N_z/N_m}$. The $1/\sqrt{N_m}$ prefactor will drop out once performing the integration over the magnon Brillouin space $\mathbf{k}$, and thus the coupling strength scales as the square root of the number of layers, $\sqrt{N_z}$. After inserting the Bogoliubov transformation, Eq. (A8), we obtain the coupling Hamiltonian Eq. (7).

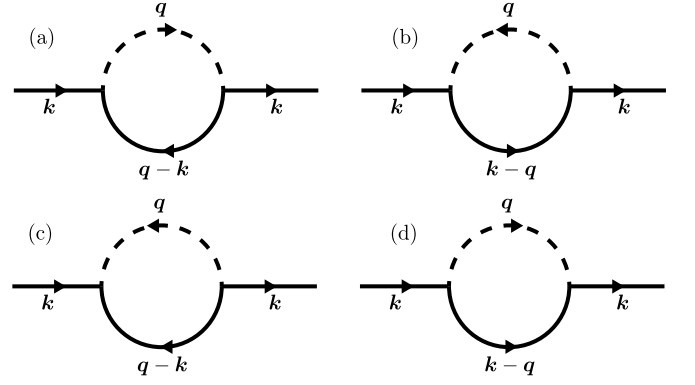


FIG. 5. Feynmann diagrams contributing to the magnon lifetime up to second order. (a) and (b) circle diagrams, (c) backward diagram, (d) forward diagram. At zero temperature only the forward diagram remains.

Appendix C: Full self-energy diagrams

In addition to the two plasmon self-energy diagrams in Fig. 3, the magnons will also gain a self energy represented by the diagrams in Fig. 5. To be fully consistent, these magnon self-energies would have to be solved in tandem with the plasmon self-energies. In particular, a finite imaginary part of the magnon self-energy would cut-off the singularity at the upper edge of the spin-zero two-magnon continuum. Note that the magnons already have a finite lifetime, due to relaxation processes and higher-order magnon-magnon interactions, and thus the singularity is already cut.

To further analyze these self-energies, we first note that the circle and backward diagram (a-c) vanish at zero temperature. We are thus left with the forward diagram in (d), which is given by

$$\Sigma(\mathbf{k}, \epsilon)^\sigma = \frac{1}{N_p} \sum_q \frac{1}{2} \frac{|\mathbf{E}_q \cdot \mathbf{W}_{q,\mathbf{k}}^\sigma|^2}{\epsilon^\sigma - \epsilon_{\mathbf{k}-q}^\sigma - \mathcal{E}_q + i0^+}, \quad (\text{C1})$$

where $\mathbf{W}_{q,\mathbf{k}}^\sigma$ is an interaction vertex, which can be found from Eq. (7). As a first step, we assume the magnon energy to be on-shell, i.e., $\epsilon^\sigma = \epsilon_{\mathbf{k}}^\sigma$, such that we can write the imaginary part of this self energy as

$$\text{Im } \Sigma(\mathbf{k})^\sigma = \frac{1}{N_p} \sum_q \frac{1}{2} |\mathbf{E}_q \cdot \mathbf{W}_{q,\mathbf{k}}^\sigma|^2 \delta(\epsilon_{\mathbf{k}}^\sigma - \epsilon_{\mathbf{k}-q}^\sigma - \mathcal{E}_q). \quad (\text{C2})$$

Since we have that $q \ll k$, we expand the magnon energies in small q to find $\epsilon_{\mathbf{k}-q}^\sigma \approx \epsilon_{\mathbf{k}}^\sigma - \mathbf{q} \cdot \mathbf{v}_{\mathbf{k}}^\sigma$, where $\mathbf{v}_{\mathbf{k}}^\sigma \equiv \nabla_{\mathbf{k}} \epsilon_{\mathbf{k}}^\sigma$ is the group velocity. The delta function then becomes $\delta(\mathbf{q} \cdot \mathbf{v}_{\mathbf{k}}^\sigma - \mathcal{E}_q)$, implying that in order for the imaginary part of the above self energy to be non-zero at any part of the magnon Brillouin zone, one requires

$$\max_{\text{BZ}} |\mathbf{v}_{\mathbf{k}}^\sigma| > \mathcal{E}_q/q \quad (\text{C3})$$

However, given the steep plasmon dispersion compared to the magnon dispersion, this is not possible for realistic param-

eters.² We thus conclude that there is no imaginary part of the above self energy which could renormalize the singularity at the upper edge of the spin-zero two-magnon continuum.

Finally, we note that one could also consider the quadratic magneto-optic interaction, which for simple cubic lattices gives the off-resonant interaction [41] **PG: is it** $(\hat{\mathbf{E}}_i \cdot \mathbf{r}_{ij})(\hat{\mathbf{E}}_j^\dagger \cdot \mathbf{r}_{ij})$ or $(\hat{\mathbf{E}}_i \cdot \mathbf{r}_{ij})(\hat{\mathbf{E}}_i^\dagger \cdot \mathbf{r}_{ij})$?

$$\hat{H}_q = - \sum_{ij} B(\hat{\mathbf{E}}_i \cdot \mathbf{r}_{ij})(\hat{\mathbf{E}}_j^\dagger \cdot \mathbf{r}_{ij})\hat{S}_i \cdot \hat{S}_j. \quad (\text{C4})$$

Here B is a phenomenological coupling strength. Firstly, we note that this interaction is quadratic in electric field, which is typically small and thus we expect this interaction to be weaker. A introducing the Holstein-Primakoff operators, Fourier transforming and applying the Bogoliboiv transformation, the interaction becomes

$$\hat{H}_q = \frac{S}{N_m} \sum_{\mathbf{k} \mathbf{q} \mathbf{q}'} (\mathbf{E}_q \cdot \boldsymbol{\Pi}_{\mathbf{q}+\mathbf{k}})(\mathbf{E}_{\mathbf{q}'}^\dagger \cdot \boldsymbol{\Pi}_{\mathbf{q}+\mathbf{k}})\hat{\phi}_{\mathbf{q}}^\dagger \hat{\phi}_{\mathbf{q}'} \times \\ (u_{\mathbf{k}} u_{\mathbf{q}'-\mathbf{q}-\mathbf{k}} \alpha_{\mathbf{k}} \beta_{\mathbf{q}'-\mathbf{q}-\mathbf{k}} + v_{\mathbf{k}} v_{\mathbf{q}'-\mathbf{q}-\mathbf{k}} \alpha_{-\mathbf{k}}^\dagger \beta_{-\mathbf{q}'+\mathbf{q}+\mathbf{k}}^\dagger) + (\mathbf{q} \leftrightarrow \mathbf{q}'), \quad (\text{C5})$$

where we have only kept the interactions that will generate a finite self energy at zero temperature. Here, $\boldsymbol{\Pi}_{\mathbf{k}} = B \sum_{ij} e^{-i\mathbf{r}_{ij} \cdot \mathbf{k}} \mathbf{r}_{ij}$ is the Fourier transform of the nearest-neighbor vectors $\mathbf{r}_{ij}$. We now construct the (Matsubara) self energy

$$\Sigma_q(\epsilon) = \frac{N_z^2}{N_m^2} \sum_{\sigma\sigma'} \sum_{\mathbf{k}\mathbf{q}'} |(\mathbf{E}_q \cdot \boldsymbol{\mathcal{Y}}_{\mathbf{q},\mathbf{k}}^{\sigma\sigma'}) (\mathbf{E}_{\mathbf{q}'}^* \cdot \boldsymbol{\mathcal{Y}}_{\mathbf{q},\mathbf{k}}^{\sigma\sigma'})|^2 \times \\ \mathcal{G}^p(q') \mathcal{G}_\sigma^m(k) \mathcal{G}_{\sigma'}^m(q - q' - k), \quad (\text{C6})$$

where $\boldsymbol{\mathcal{Y}}_{\mathbf{q},\mathbf{k}}^{\sigma\sigma'}$ is the interaction vertex and we used the short-hand notation $\mathbf{k} \equiv (\mathbf{k}, i\omega_n)$. We show the corresponding Feynmann diagram in Fig. 6.

Following the same arguments as in the main text, we expect that the self energy in Fig. 6 is only significant for the excitation of two Brillouin zone edge magnons, such that $\mathbf{q} = \mathbf{q}'$. As a first approximation this means we can insert $\delta_{\mathbf{q},\mathbf{q}'}$ and perform the summation over $\mathbf{q}'$ to obtain

$$\Sigma_q(\epsilon) \approx \frac{N_z^2}{N_m^2} \sum_{\sigma\sigma'} \sum_{\mathbf{k}} |\mathbf{E}_q \cdot \boldsymbol{\mathcal{Y}}_{\mathbf{q},\mathbf{k}}^{\sigma\sigma'}|^4 \mathcal{G}^p(q) \mathcal{G}_\sigma^m(k) \mathcal{G}_{\sigma'}^m(-k). \quad (\text{C7})$$

Comparing with the forward diagram we consider in the main text [Eq. (12)] we can conclude that this self-energy diagram has a significantly reduced phase space available and is suppressed by N_z/N_m . We therefore expect that this self energy

is much weaker. In addition, this self energy is quartic in the electric field, which will suppress its strength further.

² In particular, note that $\mathcal{E}_q \propto \sqrt{q}$ and thus $\mathcal{E}_q/q \propto q^{-1/2}$.

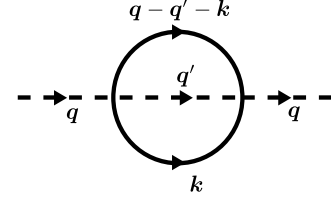


FIG. 6. Feynmann diagram of the plasmon energy resulting from the quadratic magneto-optical interaction.

Appendix D: Plasmon dispersion and generated electric field

Here we reproduce for completeness the dispersion and generated electric field of plasmons, following Ferreira *et al.* [18]. We consider a 2DEG surrounded by a dielectric medium with dielectric constant ϵ . The plasmon dispersion then becomes

$$\omega_{q,p}^2 = \frac{1}{2} \frac{D}{2\epsilon_0\epsilon} \sqrt{4q^2 + \left(\frac{D}{2\epsilon_0 c^2}\right)^2} - \frac{1}{2} \left(\frac{D}{2\epsilon_0 \sqrt{\epsilon} c}\right)^2, \quad (\text{D1})$$

Here, D is the Drude weight, which for simplicity we take to be $D = e^2 E_F / (\pi \hbar^2)$, as is the case for graphene. The electric field can then be found as

$$\mathbf{E}(\mathbf{r}) = i \sqrt{\frac{\hbar \omega_{q,p}}{2A\epsilon_0 N_q}} e^{i\mathbf{q} \cdot \mathbf{r}} \left(i\hat{\mathbf{q}} - \frac{q}{\kappa_q} \text{sign } z\hat{\mathbf{z}} \right) e^{-z/\lambda_q}, \quad (\text{D2})$$

where

$$N_q = \frac{\epsilon}{\kappa_q^3} (\kappa_q^2 + q^2) \quad (\text{D3})$$

is the mode length and

$$\kappa_q = \sqrt{q^2 - \omega_{q,p}^2 \epsilon / c^2} \quad (\text{D4})$$

is the out-of-plane momentum. Finally, $\lambda_q = \kappa_q^{-1}$ is the decay length along the z -axis. Throughout this work, we take $E_F = 100$ meV and a typical dielectric constant for insulating compounds, $\epsilon = 10$, which holds for example for MnF_2 [42].

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